

ABG Pattern Finder

Use pH, PaCO2, and bicarbonate direction to identify the primary pattern; clinical context determines the cause.

Reference frame - pH about 7.35-7.45. - PaCO2 about 35-45 mm Hg. - HCO3 about 22-26 mEq/L. - Use the reporting laboratory range.	Direction rules - Respiratory: pH and PaCO2 move opposite. - Metabolic: pH and HCO3 move together. - A value moving in the other system may show compensation.
Clinical anchors - Ventilation problems change CO2. - Renal/GI/metabolic problems change bicarbonate or acids. - Airway and breathing instability come before worksheet interpretation.	

Reviewed 2026-07-13 - Source: MedlinePlus, U.S. National Library of Medicine - <https://medlineplus.gov/lab-tests/electrolyte-panel/>
Study aid only. Verify current references, instructor guidance, scope, orders, and facility policy.

Common Reversal and Rescue Associations

Exam-focused associations only; real use depends on the agent, timing, client, protocol, and prescriber.

Frequently tested pairs - Opioids - naloxone. - Benzodiazepines - flumazenil in selected situations. - Heparin - protamine. - Warfarin - vitamin K; urgent reversal may require additional products. - Acetaminophen - N-acetylcysteine.	Important cautions - Reversal can uncover pain, withdrawal, seizures, thrombosis, or recurrent toxicity. - Some antidotes have narrow indications and monitoring needs. - Call poison control/rapid response/emergency services per setting.
Exam approach - Identify the agent. - Protect airway, breathing, and circulation. - Stop exposure and escalate. - Then connect the approved antidote and monitoring.	

Reviewed 2026-07-13 - Source: U.S. Food and Drug Administration - <https://www.fda.gov/drugs/drug-safety-and-availability/medication-errors-related-cder-regulated-drug-products>
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Common Laboratory Values

High-yield adult reference ranges with an explicit reminder that laboratories and client factors vary.

CBC landmarks - WBC, hemoglobin, hematocrit, and platelet ranges vary by lab, age, sex, pregnancy, and altitude. - Interpret trends and symptoms, not a memorized number alone. - Verify the exact report range before acting.	Chemistry landmarks - Sodium, potassium, glucose, BUN, creatinine, and bicarbonate help frame fluid, renal, metabolic, and acid-base status. - Hemolysis can falsely change some results, especially potassium. - Critical values require read-back and policy-based escalation.
Memory frame - CBC = cells. - BMP/CMP = chemistry and organ clues. - Coagulation tests = clotting pathway and medication monitoring context.	

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Therapeutic Communication + ADPIE

Combine a safe conversation style with the nursing-process and clinical-judgment cycle.

Therapeutic moves - Open-ended invitation, reflection, clarification, silence, validation, and focused exploration. - Acknowledge emotion before giving information. - Use teach-back without testing or shaming.	Avoid - False reassurance, "why" questions that sound blaming, advice-giving, topic changes, arguing, and promises you cannot keep. - Do not reinforce a delusion; respond to the feeling and present reality. - Do not keep safety threats secret.
ADPIE + judgment - Assess/recognize cues. - Diagnose or analyze cues and prioritize hypotheses. - Plan/generate solutions. - Implement/take action. - Evaluate outcomes and revise.	

Reviewed 2026-07-13 - Source: Agency for Healthcare Research and Quality - <https://www.ahrq.gov/teamstepps-program/resources/modules/index.html>
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Cranial Nerves

A function-first map for focused neurologic assessment.

Sensory - I smell. - II vision and visual fields. - VIII hearing and balance.	Eye and face - III, IV, VI eye movement; III also pupil response and eyelid elevation. - V facial sensation and chewing. - VII facial movement and taste.
Swallow, voice, and movement - IX/X swallow, palate, voice, gag context. - XI shoulder shrug and head turn. - XII tongue movement. - Compare sides and connect findings to the overall neuro picture.	

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Electrolyte Imbalances

Notice the direction, connect it to symptoms, and protect the heart and brain while escalating.

Potassium - Low or high potassium can disturb cardiac rhythm and muscle function. - Check renal status, GI losses, medications, and ECG context. - Never treat a number without verifying the result and order.	Sodium - Sodium disorders commonly show neurologic changes because water shifts affect the brain. - Track fluid balance, weight, intake/output, and trend. - Correction rate is a provider-managed safety issue; avoid rapid unplanned changes.
Calcium and magnesium - Low levels can increase neuromuscular irritability. - High levels can reduce reflexes or alertness. - Watch respiratory, cardiac, and seizure safety cues.	

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Glasgow Coma Scale and Neuro Trend

Trend eye, verbal, and motor responses using the same stimulus and documentation method.

Three domains - Eye opening: 1-4. - Verbal response: 1-5. - Motor response: 1-6. - Total score: 3-15, but component scores tell more.	Document clearly - Record E, V, and M components, total, time, and stimulus. - Note intubation, sedation, language, hearing, paralysis, and baseline limits. - Avoid vague terms such as "stable neuro."
Escalate change - A declining trend, new pupil difference, new weakness, seizure, or acute behavior change needs prompt assessment and escalation. - Protect airway and injury risk. - Check glucose and reversible factors per protocol.	

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Heart and Lung Sound Clues

Use location, timing, quality, symmetry, symptoms, and trend together.

Heart landmarks - S1 marks closure at the start of systole; S2 marks closure at the end. - Extra sounds or murmurs require context and trained assessment. - Apical rate and rhythm help verify an irregular peripheral pulse.	Lung landmarks - Crackles: discontinuous popping, often linked with fluid or reopening airways. - Wheezes: musical narrowing clue. - Stridor: upper-airway emergency clue. - Diminished/absent: consider poor air movement, obstruction, or other urgent causes.
Assess before naming - Compare side to side. - Note work of breathing, speech, color, cough, and oxygen data. - Escalate acute asymmetry, stridor, silence with distress, or rapid deterioration.	

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Injection Site Decision Guide

Site and technique depend on route, medication, volume, age, tissue, device, and policy.

Subcutaneous - Use approved fatty-tissue sites. - Rotate sites for repeated injections. - Choose angle and skin fold for needle length and client tissue.	Intramuscular - Use an approved landmarked muscle and needle selected for client and medication. - Avoid injured, infected, scarred, or contraindicated areas. - Follow specific product guidance for site and technique.
Always - Verify route and formulation. - Use a safety-engineered needle/device. - Never recap; discard directly into sharps. - Check current school/facility policy.	

Reviewed 2026-07-13 - Source: AHRQ Patient Safety Network - <https://psnet.ahrq.gov/primer/medication-administration-errors>
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Insulin Families and Safety

A class-level memory aid; always verify the exact product label, meal plan, order, and facility policy.

Timing families - Rapid-acting: begins quickly and is linked closely to food availability. - Short-acting: slower onset and a more visible peak. - Intermediate: cloudier suspension for some products; has a meaningful peak. - Long/ultra-long: basal coverage with little or no pronounced peak.	Safety sequence - Check glucose, symptoms, nutrition status, exact product, dose, and timing. - Use the correct device and independent check when required. - Know the local hypoglycemia plan before giving insulin.
Hypoglycemia clues - Sweating, tremor, hunger, palpitations, behavior change, confusion, or reduced consciousness. - Confirm when possible without delaying emergency care. - Treat and recheck only according to the approved protocol.	

Isolation Precautions

Start with Standard Precautions, then add transmission-based measures from the current sign and policy.

Contact - Gown and gloves when room-entry policy/interaction requires them. - Dedicate or disinfect equipment. - Contain contaminated items and perform hand hygiene.	Droplet - Medical mask for close respiratory exposure as directed. - Mask the client for necessary transport when tolerated. - Special air handling is generally not required.
Airborne - Use an airborne infection isolation room when required. - Wear a fit-tested respirator according to policy. - Keep the door closed and limit transport.	

Reviewed 2026-07-13 - Source: Centers for Disease Control and Prevention - <https://www.cdc.gov/infection-control/hcp/isolation-precautions/precautions.html>
Study aid only. Verify current references, instructor guidance, scope, orders, and facility policy.

IV Calculation Formulas

A compact setup-and-unit checklist for pump, gravity, and completion-time problems.

Pump - $\text{mL/hr} = \text{total mL} / \text{hours}$. - Convert minutes to a fraction of an hour before calculating. - $\text{Completion hours} = \text{remaining mL} / \text{mL/hr}$.	Gravity - $\text{gtt/min} = (\text{mL} \times \text{drop factor}) / \text{total minutes}$. - Drop factor comes from the tubing package. - Round to whole drops unless instructed otherwise.
Safety check - Write units at every step. - Estimate before calculating. - Trace the line, verify concentration, pump library, and independent-check requirements.	

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Maternal and Fetal Priority Cues

A study sheet for recognizing urgent changes; follow current obstetric protocols and instructor guidance.

Prenatal warning cues - Heavy bleeding, severe persistent headache, vision changes, seizure, chest pain, severe dyspnea, or markedly reduced fetal movement warrant prompt evaluation. - Check gestational context and baseline. - Avoid giving individualized advice from a study aid.	Labor framework - Assess fetal response, contraction pattern, maternal vital signs, pain/coping, membranes/fluid, and labor progress. - Reposition and escalation decisions follow the specific tracing, order, and protocol. - Communicate trends, not isolated data.
Postpartum/newborn - Heavy bleeding, uterine tone change, shock cues, severe pain, fever, or new neurologic symptoms are urgent. - For the newborn, airway/breathing, temperature, feeding, glucose risk, color, and behavior are key. - Use two-identifier and security procedures.	

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Medication Name Patterns

Suffixes are study clues, not proof of a drug class or a substitute for checking the label.

Cardiovascular clues -pril often signals an ACE inhibitor. -sartan often signals an ARB. -olol often signals a beta blocker. -dipine often signals a dihydropyridine calcium-channel blocker.	Other common clues -statin: lipid-lowering statin. -prazole: proton-pump inhibitor. -cillin: penicillin-family antibiotic. -gliptin: DPP-4 diabetes medication.
Use the clue safely - Confirm generic name, indication, route, dose, and allergies. - Look-alike names can cross class boundaries. - Use an approved drug reference for monitoring and teaching.	

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Oxygen Devices and Checks

Know the device family, assess the client, and never substitute a memorized flow range for the order and policy.

Common families - Nasal cannula: low-flow, allows talking/eating. - Simple or reservoir masks: fit and minimum-flow safety matter. - Venturi-style mask: controlled oxygen concentration. - High-flow systems: heated/humidified support with specific equipment.	Nursing checks - Airway, work of breathing, speech, mental status, color, and trend. - Correct device, fit, setting, tubing, humidification, and skin protection. - Fire safety and no petroleum-based products.
Escalate - New severe distress, cyanosis, reduced consciousness, silent chest, stridor, or unexpected deterioration. - Do not delay help while troubleshooting equipment. - Follow the prescribed target and protocol.	

Pain Assessment Tools

Match the tool to age, communication ability, cognition, and setting; reassessment completes the cycle.

Self-report tools - Numeric rating scale for clients who can quantify pain. - Verbal descriptor scale when words fit better than numbers. - Faces scales can support selected children and adults.	Behavior tools - Use a validated behavioral scale when self-report is not possible. - Behavior does not prove absence or presence by itself. - Ask family about baseline expressions when appropriate.
PQRST plus function - Provokes/palliates, Quality, Region/radiation, Severity, Timing. - Add functional goal, associated symptoms, and safety risks. - Reassess after every intervention at the required interval.	

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Pediatric Milestone Study Map

Milestones are ranges and patterns; screening and caregiver concerns matter more than a rigid birthday cutoff.

Infancy themes - Rapid motor progression from head control toward sitting and mobility. - Social smile, babble, and attachment behaviors emerge over time. - Safe sleep, feeding, growth, and immunization teaching are central.	Toddler/preschool themes - Language expands, play becomes more interactive, and independence grows. - Expect regression during illness or stress. - Use simple choices and concrete explanations.
School-age/adolescent themes - Peers, privacy, identity, and body changes gain importance. - Include the child in teaching at the right developmental level. - Report lost skills or significant delay concerns for evaluation.	

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Prioritization and Delegation

A repeatable decision ladder for exam items and clinical communication.

Priority ladder - Immediate threats to airway, breathing, circulation, safety, or acute neurologic status. - Unstable before stable; acute/unexpected before chronic/expected. - Actual problems often before risk, unless the risk is immediate and catastrophic.	Delegation check - Right task, circumstance, person, direction/communication, supervision/evaluation. - Assessment, nursing judgment, teaching, and evaluation generally remain with the licensed nurse at the appropriate level. - Scope varies by role, state, and facility.
Question clue - Ask what can harm the client first. - Choose the least restrictive effective safety action. - If data are missing, assess before intervening unless delay is dangerous.	

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Cardiac Rhythm First Look

A safe rhythm-study sequence: client first, then rate, regularity, waves, intervals, and pattern.

Six-step strip scan - Assess the client and pulse. - Estimate rate. - Check regularity. - Look for P waves. - Measure PR and QRS. - Name the pattern only after the evidence.	High-priority clues - No pulse changes the response completely. - New chest pain, hypotension, altered mental status, shock, or severe dyspnea signal instability. - Artifact can imitate a lethal rhythm; check leads and client without delaying emergency response.
Memory line - Treat the client, not the monitor. - A rhythm name is not a complete assessment. - Follow current resuscitation training and facility algorithms.	

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Adult Vital Signs at a Glance

Resting adult reference points plus the assessment cues that matter more than one isolated number.

Common resting reference points - Pulse: 60-100/min - Respirations: 12-18/min - Blood pressure: roughly 90/60 to below 120/80 mm Hg - Temperature varies by route, time, and person; trend against baseline	Check the context - Age, activity, pain, anxiety, medications, posture, and illness affect readings. - Choose the correct cuff and site. - Recheck an unexpected value manually when appropriate.
Escalate the pattern - New symptoms plus an abnormal trend outrank a lone number. - Airway, breathing, circulation, mental-status, and perfusion changes require prompt attention. - Follow course and facility notification parameters.	

Reviewed 2026-07-13 - Source: MedlinePlus, U.S. National Library of Medicine - <https://medlineplus.gov/ency/article/002341.htm>
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